

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	GAUTHAM G	Reg. No.:	192524371
Section / Batch:	SLOT-B	Date:	22/07/2026

Code of Conduct

I am GAUTHAM G (192524371)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rule will result in disciplinary action.

Signature of the StudentGAUTHAM G

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11- 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Accuracy					
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO3_AT3_MCQ OPEN

Multiple Choice Questions

0/25 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →

Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

```
high = mid;
}
return -1;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[],int n,int key){
2     int low=0,high=n-1;
3     while(low<=high){
4         int mid=(low+high)/2;
5         if(arr[mid]==key)
6             return mid;
7         else if(arr[mid]<key)
8             low=mid+1;
9         else
10            high=mid-1;
11    }
12 }
13 return -1;
```

Test Input

```
5
10 20 30 40 50
40
```

Output

```
Compilation Error:
C:/CCPPComp/CCPP1/bin/./lib/gcc/
x86_64-w64-
mingw32/10.3.0/./././././x86_64-
-w64-mingw32/bin/ld.exe:
C:/CCPPComp/CCPP1/bin/./lib/gcc/
```

🛠 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (expression[i] != '\0') {
2     if (isdigit(expression[i]))
3         push(expression[i] - '0');
4     else {
5         int op2 = pop();
6         int op1 = pop();
7         int result;
8         switch (expression[i]) {
9             case '+': result = op1 + op2; break;
10            case '-': result = op1 - op2; break;
11            case '*': result = op1 * op2; break;
12            case '/': result = op1 / op2; break;
12        }
13        push(result);
14    }
15    i++;
16 }
17 }
```

Test Input

231*+9-

Output

-4

(Postfix evaluation result)



Run



Save



Submit Fix



Search

ENG
IN07:52 PM
29-07-2025